HACK-A-ROO 2026 PITCH-TO-BUILD TRACK

Labor & Delivery Severe Hypertension Alert System

Real-Time Protocol Tracking, Bedside Countdown, and Traceable EMR Synthesis for Maternal Care

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🚨 1. Problem Statement & Clinical Urgency

> **Severe Maternal Morbidity:** Severe hypertension in pregnancy accounts for >9% of maternal mortality in the US and is a leading preventable cause of stroke, eclampsia, and ICU admission.

The Bedside Dilemma: In busy Labor & Delivery units, bedside nurses face high cognitive load, paper flowsheets, and constant alarms. Relying on *human memory* to track repeat measurement windows frequently leads to delayed reassessments and missed medication deadlines.

- **Severe Range Blood Pressure:** Systolic ≥ 160 mmHg OR Diastolic ≥ 110 mmHg.
- **The 15–60 Minute Recheck Window:** Guidelines require a timely repeat blood pressure to confirm persistent severe hypertension.
- **The 60-Minute Treatment Target:** Standard antihypertensives (IV labetalol, IV hydralazine, or oral nifedipine) must be administered within **60 minutes of onset** (measured from the *first* severe reading, not confirmation).

📊 2. Datasets & Synthetic Stream Modeling

Per clinical research safety rules, the system operates strictly on **open benchmark data and reproducible synthetic streams** (Zero real patient PHI):

> **Open Maternal Benchmark (UCI Repository):** 1,014 de-identified clinical records across Age, Systolic BP, Diastolic BP, Glucose, Temp, and Heart Rate (Ahmed et al., 2020; DOI: 10.24432/C5DPSD, CC BY 4.0).

8-Scenario Synthetic Patient Generator (`generator.py`):

- **Severe Trajectories:** `persistent_severe`, `nonpersistent_severe`, `recurrent_severe`.
- **Workflow Delays:** `delayed_or_missing_repeat` (explicit failure modes: +16m delay, +30m missing recheck).
- **Control Baselines:** `non_hypertensive` (<130/<80 mmHg), `mild_hypertension` (140–159/90–109 mmHg).

1,014	8 Scenarios
UCI BENCHMARK RECORDS	SYNTHETIC PATIENT PROFILES

🎯 Project Objectives

- **Deterministic FSM:** Eliminate reliance on bedside memory with an SMFM 2022-compliant state machine. Zero black-box AI in the safety loop.
- **Bedside Visual Timelines:** Live 15-minute countdowns and unit multi-bed monitoring for immediate situational awareness.
- **Traceable EMR Synthesis:** Generative AI for nursing documentation with 100% token mathematical verification against system events.

📐 3. System Architecture & Flow

Rigorous architectural decoupling: safety-critical alerting remains 100% deterministic, while generative AI is confined to post-event documentation.

⏱️ 4. SMFM 2022 State Machine Engine

Implements all rules from SMFM Figure 1 & Table 1 in `episodes.py`:

- **`EPISODE_OPENED`:** Fires immediately on severe reading. The 60-minute medication clock anchors to onset time (T_{onset}).
- **`REPEAT_DUE`:** Prompt scheduled for exactly 15 minutes after onset (T_{onset} + 15 min).
- **`CONFIRMED`:** Fires upon repeat severe reading in 15–60m window, *OR* via tick() after 15m if no normal check is documented (missing recheck rule).
- **`ESCALATED`:** Fires if attending provider notification is unacknowledged for > 10 min.
- **`EPISODE_REOPENED`:** A severe reading within 60m folds back into the existing episode.

🛡️ 5. Traceable EMR AI Documentation

Generative AI drafts clinical event timelines, protected by strict mathematical guardrails:

> **The Traceability Rule:** Every timestamp (e.g., 08:15), blood pressure reading (e.g., 168/112 mmHg), and bed identifier (e.g., Bed-04) in the draft must map back to an immutable system Event. Invented values are blocked.

- **Zero Diagnosis:** Prohibited from stating clinical diagnoses (e.g. preeclampsia).
- **Zero Treatment Orders:** Strictly documents administered doses; never recommends treatment.
- **Nurse-in-the-Loop:** Must be reviewed and verified by the bedside nurse before EMR commit.

📈 6. Paired Workflow Simulation Results

Monte Carlo paired simulation comparing standard baseline practice vs. alert-assisted care over **80 synthetic patients**:

Clinical Workflow Milestone	Baseline	Alert Tool	Net Impact
Nurse Recognition Latency	7.00 min	2.75 min	**4.25 min saved**
Reassessment Lateness	8.85 min	4.50 min	**4.35 min saved**
Provider Notification Delay	12.45 min	5.00 min	**7.45 min saved**
Time to Medication (from onset)	73.35 min	65.90 min	**7.45 min saved**
Treated ≤ 60 Min Target	5 / 20 (25%)	8 / 20 (40%)	**+60.0% Relative Gain**

-7.45m	+60.0%
MEDICATION DELIVERY SPEEDUP	TARGET ADHERENCE ADOPTER GAIN

🌊 7. Sensitivity Analysis & Verification

Varying nurse prompt adoption from 0% to 100% demonstrates robust latency reduction:



Test Suite Integrity: 238 passing tests across 16 test modules with 0 failures and 0 xfails. 100% verification accuracy on invented/hallucinated token detection.

☀️ 8. Key Contributions & Future Work

- **Clinical Guideline Fidelity:** First open-source platform enforcing exact SMFM 2022 timing metrics without heuristic drift.
- **Safety-First Architecture:** Clean separation between life-critical deterministic alerts and AI-assisted documentation.
- **Future Work:** Direct HL7/FHIR monitor telemetry ingestion, integration into Epic/Cerner EHR, and clinical evaluation in maternal mannequin simulation labs.